The complexity of multistage minimum utility cost

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Abstract. Letsios et al. ask whether minimum utility cost in a multistage heat-exchanger superstructure is polynomial-time solvable or NP-hard. We distinguish their displayed journal equations, which omit temperature-approach constraints, from two physical completions: approaches at the mixed stage boundaries and at the actual counter-current exchanger endpoints. The displayed equations have a closed-form optimum. Both physical completions are strongly NP-hard, even with exactly two stages, all stream pairs permitted, unit utility prices and the question of zero utility. A quadratic temperature identity forces every mixing variance and every exchanger approach slack to vanish in the reduction. This makes each hot stream choose a bin in a 3-Partition instance. Padding proves hardness for every fixed number of stages at least two. With one stage the counter-current version is an LP, whereas the boundary version is strongly NP-complete. The boundary decision problem is in NP; the counter-current problem is in the existential theory of the reals, and we do not establish its general membership in NP. Exact symbolic and rational checks corroborate the proofs; floating-point optimization is not used as a certificate.

The question and the model correction

Dimitrios Letsios, Radu Baltean-Lugojan, Francesco Ceccon, Miten Mistry, Johannes Wiebe and Ruth Misener, *Approximation algorithms for process systems engineering*, *Computers & Chemical Engineering* 132 (2020), 106599, DOI 10.1016/j.compchemeng.2019.106599, state in Section 6.4, journal page 10 [1]:

> *"The complexity of the multistage minimum utility cost problem is an intriguing open question."*

The supplied journal PDF was checked visually. Its inequalities (5j,k) have the correct decreasing signs, unlike arXiv v1. Equation (5m) still indexes cold streams by the hot set; we necessarily read that index as the cold set. No constraint in the display compares a hot temperature to a cold temperature. Thus a physical interpretation requires an explicit additional assumption. Our answer is for the two completions defined below, not a claim that the missing constraints were printed in the source.

Theorem 1. For each fixed integer $r \geq 2$, deciding whether the minimum utility cost is zero in the r-stage counter-current model C is strongly NP-hard. This holds with positive integer stream capacities, all hot–cold pairs available, both utility prices equal to one, and a minimum approach fixed to either zero or any prescribed nonnegative rational constant. Temperatures can be positive integers when the fixed approach is an integer, and have constant denominators for any other fixed rational approach. The same restrictions give strong NP-hardness for the stage-boundary model B. Consequently, both problems are strongly NP-hard when the stage count is part of the input, in unary or binary.

For B, the decision problem with any rational cost threshold is in NP and hence strongly NP-complete. For C it is in $\exists\mathbb{R}$; no general NP membership assertion is

made. For one stage, C is polynomial-time solvable and B is strongly NP-complete.

1 Exact formulation and input encoding

Let I, J be nonempty hot and cold stream sets, of sizes n, s. The given positive heat-capacity flow rates are F_i, G_j. Given temperatures satisfy $H_i \geq \underline{H}_i \geq 0$ and $\overline{C}_j \geq C_j \geq 0$: hot stream i must cool from H_i to $\underline{H}_i$, and cold stream j must heat from C_j to $\overline{C}_j$. The stage count is $\ell \geq 1$. Unless explicitly fixed, it is an input integer. All numerical data, including the common utility prices $c_{\mathrm{CU}}, c_{\mathrm{HU}} \geq 0$, minimum approach $\delta \geq 0$, and decision threshold, are binary-encoded rationals. Our main reductions fix ℓ and both prices; they do not encode complexity in a large rational denominator or in the number of stages.

Index boundaries by $k = 0, \ldots, \ell$. Hot streams move from boundary 0 to ℓ, and cold streams move in the opposite direction. Their mixed temperatures are h_{ik}, c_{jk}. Stage $k \in \{1, \ldots, \ell\}$ has variables f_{ijk}, g_{ijk} for the hot and cold branch capacities, a_{ijk}, b_{ijk} for their respective outlet temperatures, and q_{ijk} for exchanged heat. These are real decision variables, all nonnegative. Branches split from one mixed inlet and mix again before the next stage; there is no cross-stream material mixing. All pairs (i, j) are available in every stage. The base equations, denoted N, are the journal's (5), with only the necessary cold-index normalization and notation changes:

$$h_{i0} = H_i, \qquad h_{i\ell} \geq \underline{H}_i, \qquad c_{j\ell} = C_j, \quad c_{j0} \leq \overline{C}_j, \tag{2}$$

$$h_{i,k-1} \geq h_{ik}, \qquad c_{j,k-1} \geq c_{jk}, \tag{3}$$

$$\sum_j f_{ijk} = F_i, \qquad \sum_i g_{ijk} = G_j, \tag{4}$$

$$q_{ijk} = f_{ijk}(h_{i,k-1} - a_{ijk}) = g_{ijk}(b_{ijk} - c_{jk}), \tag{5}$$

$$F_i h_{ik} = \sum_j f_{ijk} a_{ijk}, \qquad G_j c_{j,k-1} = \sum_i g_{ijk} b_{ijk}. \tag{6}$$

The nonnegative end utilities and objective are

$$U_i = F_i(h_{i\ell} - \underline{H}_i), \qquad V_j = G_j(\overline{C}_j - c_{j0}), \qquad \min\ c_{\mathrm{CU}} \sum_i U_i + c_{\mathrm{HU}} \sum_j V_j. \tag{7}$$

Zero-heat branches may carry bypass flow. There is no minimum positive heat duty, match-count restriction, exchanger-area bound, pressure-drop term, capital charge, or intermediate utility. End utilities are available without capacity restrictions. These assumptions isolate the source's utility-cost problem from total annual cost and minimum matches.

Stages here are ordered structural positions, not prescribed temperature intervals. Their boundary temperatures are optimized. In particular, the fixed temperature-interval transshipment model discussed separately in the source is not being substituted for this model.

1.1 The two physical completions

Model C adds, for every $q_{ijk} > 0$, the counter-current endpoint constraints

$$h_{i,k-1} \geq b_{ijk} + \delta, \qquad a_{ijk} \geq c_{jk} + \delta. \tag{8}$$

Model B instead adds, for every $q_{ijk} > 0$, the stage-boundary constraints

$$h_{i,k-1} \geq c_{j,k-1} + \delta, \qquad h_{ik} \geq c_{jk} + \delta. \tag{9}$$

An implication is imposed only on heat-carrying matches; it does not forbid unheated bypass branches between temperature-incompatible streams. Yee–Grossmann SYNHEAT uses the stage-boundary version. The public GAMS implementation explicitly activates both boundary comparisons with its match binary and sets the approach variable's lower bound to the prescribed minimum [2] [3]. For non-isothermal mixing the actual exchanger outlets are the relevant endpoints. The contemporary OpenHENS implementation displays exactly those two branch comparisons, alongside its isothermal boundary alternative [4]. We use these sources to fix the approach constraints, not their additional area objectives or design restrictions.

Adding δ to every cold temperature transforms either set of approach constraints to its zero-approach version and leaves the heat balances unchanged. This is the usual temperature shift mentioned in the source. Conversely, all our hardness instances for $\delta = 0$ become instances for any fixed $\delta > 0$ by adding δ to every hot temperature. A further common translation makes all absolute temperatures strictly positive.

Summing the branch equations gives the stage balances

$$R_{ik} := \sum_j q_{ijk} = F_i(h_{i,k-1} - h_{ik}), \qquad S_{jk} := \sum_i q_{ijk} = G_j(c_{j,k-1} - c_{jk}). \tag{10}$$

Write $E_H = \sum_i F_i(H_i - \underline{H}_i)$, $E_C = \sum_j G_j(\overline{C}_j - C_j)$ and $Q = \sum_{ijk} q_{ijk}$. The objective equals

$$c_{\mathrm{CU}} E_H + c_{\mathrm{HU}} E_C - (c_{\mathrm{CU}} + c_{\mathrm{HU}})Q. \tag{11}$$

Lemma 12 (Isothermal lifting). Any nonnegative q, h, c satisfying the endpoint and stage balances can be lifted to N. If they satisfy the active boundary comparisons, they can be lifted simultaneously to B and C.

Proof. In each stage set $a_{ijk} = h_{ik}$ and $b_{ijk} = c_{j,k-1}$. When $R_{ik} > 0$ set $f_{ijk} = F_i q_{ijk}/R_{ik}$; when $R_{ik} = 0$ use any nonnegative row summing to F_i. Likewise, when $S_{jk} > 0$ set $g_{ijk} = G_j q_{ijk}/S_{jk}$, and otherwise use any column summing to G_j. Equations (5) and (6) follow immediately from (10). On active matches, (8) becomes (9). □

Thus B and C are compared through their feasible (q, h, c) projections; an arbitrary non-isothermal assignment satisfying B need not itself satisfy C. This distinction is important in applying the next proof to B.

2 The displayed journal equations have a closed-form optimum

Theorem 13. For every $\ell \geq 1$, the optimum of N is

$$c_{\mathrm{CU}} \max(E_H - E_C, 0) + c_{\mathrm{HU}} \max(E_C - E_H, 0). \tag{14}$$

An optimal rational solution can be constructed in polynomial time, using only one nonidle stage.

Proof. The stage balances imply $Q \le \min(E_H, E_C)$, giving the lower bound in (14). Greedily transport heat from hot supplies $F_i(H_i - \underline{H}_i)$ to cold demands $G_j(\overline{C}_j - C_j)$, with no pair restrictions, until one side is exhausted. Let q_{ij} be this transportation matrix, R_i its row sums and S_j its column sums. In stage 1 set $h_{i1} = H_i - R_i/F_i$, $c_{j0} = C_j + S_j/G_j$ and $c_{j1} = C_j$. Use the lift in Lemma 12; repeat these post-stage temperatures through all later, idle stages. Then $Q = \min(E_H, E_C)$. All entries have polynomial rational bit length. A binary-encoded large ℓ is represented by this one active stage and repeated idle stages, rather than exponentially many explicit entries. □

For example, one capacity-one hot stream $1 \to 0$ and one capacity-one cold stream $10 \to 11$ have zero utility in N: the exchanger transfers one heat unit uphill in temperature. Thus (14) is a model correction, not a resolution of the intended physical question.

3 A quadratic identity that removes fractional mixing

We first take $\delta = 0$. For a single exchanger abbreviate its hot inlet and outlet to h, a and its cold outlet and inlet to b, c. Its balances imply

$$f(h^2 - a^2) - g(b^2 - c^2) = q[(h - b) + (a - c)]. \tag{15}$$

For each stage define the nonnegative mixing variances

$$W_k^H = \sum_{ij} f_{ijk}(a_{ijk} - h_{ik})^2, \qquad W_k^C = \sum_{ij} g_{ijk}(b_{ijk} - c_{j,k-1})^2. \tag{16}$$

Lemma 17 (Quadratic budget and equality case)**.** Every feasible C solution satisfies

$$\begin{aligned} &\sum_i F_i(H_i^2 - h_{i\ell}^2) - \sum_j G_j(c_{j0}^2 - C_j^2) \\ &\quad = \sum_{kij} q_{ijk}[(h_{i,k-1} - b_{ijk}) + (a_{ijk} - c_{jk})] + \sum_k (W_k^H + W_k^C) \ \ge 0. \end{aligned} \tag{18}$$

If the left side is zero, every positive-flow branch exits at its stream's mixed outlet temperature, and every active exchanger satisfies

$$h_{i,k-1} = c_{j,k-1}, \qquad h_{ik} = c_{jk}. \tag{19}$$

Proof. Weighted variance and (6) give $\sum_j f_{ijk} a_{ijk}^2 = F_i h_{ik}^2 + \sum_j f_{ijk}(a_{ijk} - h_{ik})^2$, with the analogous cold identity. Substitute these in the sum of (15). The difference of mixed hot and cold quadratic drops in each stage equals the exchanger terms plus both variances. Summing stages telescopes to (18). Every term on the right is nonnegative by (8); for $q = 0$ the exchanger term is zero regardless of unused temperatures. Zero total forces zero variances. For $q > 0$, both branch capacities are positive by (5), so $a = h_{ik}$ and $b = c_{j,k-1}$; both nonnegative approach slacks also vanish. These statements imply (19). □

The identity is algebraic; it is not an unproved appeal to entropy or an assumption of isothermal mixing. It forces that mixing property on the particular instances constructed next.

4 Strong NP-hardness with two stages

Start from 3-Partition: $3m$ positive integers $a_1, \dots, a_{3m}$ and B satisfy

$$B/4 < a_i < B/2, \qquad \sum_i a_i = mB. \tag{20}$$

The question is whether they can be partitioned into m triples of sum B. This is strongly NP-complete [5]. Set $T = m + 1$, $\ell = 2$, $\delta = 0$, both utility prices to one, and the cost threshold to zero. Introduce the following streams:

Stream	Capacity	Inlet	Target
Hot item i	a_i	T	0
Cold upper U_j, $1 \le j \le m$	B	j	T
Cold lower V_j, $1 \le j \le m$	B	0	j

There are no forbidden matches or bounds on their number. Energy and quadratic temperature budgets agree exactly:

$$E_H = E_C = mBT, \tag{21}$$

$$\sum_i a_i T^2 = \sum_{j=1}^{m} B\left[(T^2 - j^2) + j^2\right] = mBT^2. \tag{22}$$

Lemma 23. The constructed C instance has zero utility if and only if the 3-Partition instance is a YES instance. The equivalence also holds for B.

Proof. Suppose first that C has zero utility. Positive utility prices force every stream to reach its prescribed target, so (22) makes the left side of (18) zero. Every active match consequently obeys (19).

No U_j can receive heat in stage 2: its inlet there is $j > 0$, whereas every hot stream's outlet there is 0. These cannot be equal as required by (19). Hence $c_{U_j,1} = j$ and the entire heat $B(T - j)$ is received in stage 1. Similarly, no V_j can receive heat in stage 1: its outlet there is its target $j < T$, whereas every hot inlet there is T. Thus $c_{V_j,1} = j$ and its entire heat Bj is received in stage 2.

Let $x_i = h_{i1}$. If $x_i < T$, hot stream i releases positive heat in stage 1 and must meet some U_j, forcing $x_i = j$. If instead $x_i = T$, it releases positive heat in stage 2 and must meet some V_j, forcing $T = j$, impossible. Therefore every x_i belongs to $\{1, \dots, m\}$. For $x_i = j$, the only possible active partners are U_j in stage 1 and V_j in stage 2. The stage-1 balance of U_j now gives

$$(T - j) \sum_{i:x_i=j} a_i = B(T - j), \qquad \sum_{i:x_i=j} a_i = B. \tag{24}$$

The strict bounds in (20) force exactly three items in every group. No fractional assignment to distinct bins survives the equality case.

Conversely, let $S_1, \dots, S_m$ be a partition with capacity B per group. For $i \in S_j$ set its hot temperatures to $(T, j, 0)$, exchange $a_i(T - j)$ with U_j in stage 1 and $a_i j$ with V_j in stage 2. Cold temperatures are (T, j, j) for U_j and $(j, j, 0)$ for V_j. These are exact stage balances and satisfy both boundary comparisons with equality. The isothermal lift supplies every branch flow and outlet temperature, including idle cold bypasses. Thus

both B and C have zero utility. Finally, any zero-utility B solution can be lifted to a C solution with the same stage heat and temperatures by Lemma 12, proving the reverse implication for B as well. □

The transformation uses $3m$ hot streams and $2m$ cold streams; its largest temperature is $m+1$ and its capacities are a_i or B. It is polynomial and preserves polynomial bounds on numerical magnitudes, proving strong NP-hardness. On this restricted family the zero-utility decision problem is itself strongly NP-complete, since a bin partition is a polynomial witness. This last statement does not establish NP membership for arbitrary C instances.

4.1 Every fixed larger number of stages

Fix $r \geq 2$, put $p = r-2$ and $A = mB$. Raise every hot inlet to $T+p$ and add the following p cold streams, leaving U_j, V_j unchanged:

$$D_t: \quad \text{capacity } A, \quad T+t-1 \longrightarrow T+t, \qquad t = 1, \ldots, p. \tag{25}$$

Use exactly r stages. Their energy and squared-temperature increments telescope, so both equalities in (21)–(22) remain true with the new hot inlet $T+p$. In a zero-utility solution, every positive hot temperature drop must exactly coincide with an interval of some cold stream by (19). It therefore lies within that cold stream's full inlet-to-target interval. No cold interval crosses any of the boundaries $T, T+1, \ldots, T+p-1$. Each hot stream needs at least one nonzero stage in each of the p upper bands to descend to T. It needs at least two more stages to descend from T to 0, because no cold stream spans that whole interval. There are only $p+2$ stages. Hence every hot stream uses exactly one full drop in each upper band, in decreasing order, followed by exactly the two stages of the original construction. There are no idle or additional partial drops. The argument of Lemma 23 applies to these last two stages. Conversely all hot streams can traverse each added band together, supplying the capacity-A dummy cold stream, before executing the partition witness. This proves hardness for each fixed r; adding δ to hot temperatures proves it for any fixed nonnegative approach. With $\delta > 0$ the witness has genuinely positive physical endpoint differences, so the reduction does not require a zero-driving-force exchanger in physical temperature coordinates.

5 One stage distinguishes the two models

5.1 A polynomial LP for the counter-current version

For $\ell = 1$, let $D_{ij} = H_i - C_j - \delta$. Delete pairs with $D_{ij} \leq 0$. The exact feasible heat projection of C is

$$q_{ij} \geq 0, \qquad \sum_j q_{ij} \leq F_i(H_i - \underline{H}_i), \qquad \sum_i q_{ij} \leq G_j(\overline{C}_j - C_j), \tag{26}$$

$$\sum_{j:D_{ij}>0} \frac{q_{ij}}{D_{ij}} \leq F_i, \qquad \sum_{i:D_{ij}>0} \frac{q_{ij}}{D_{ij}} \leq G_j. \tag{27}$$

Indeed, (8) and (5) imply $q_{ij} \leq f_{ij} D_{ij}$ and $q_{ij} \leq g_{ij} D_{ij}$. Summing proves necessity. Conversely start both branch capacities at q_{ij}/D_{ij} on retained pairs and at zero elsewhere. Allocate unused hot capacity arbitrarily within each row and unused cold capacity within each column, independently, to meet their split sums. On positive flow define $a_{ij} = H_i - q_{ij}/f_{ij}$ and $b_{ij} = C_j + q_{ij}/g_{ij}$; for zero flow choose its inlet temperature. The capacity lower bounds guarantee (8), and the mixed temperatures and utility bounds

follow from (26). Maximize $\sum q_{ij}$ in this rational LP, or use (11) directly. This proves polynomial solvability, including construction of an optimal network.

5.2 Strong NP-hardness of one-stage boundary approaches

Use the same 3-Partition data. Give each hot stream capacity a_i and range $m \to 0$; give cold stream $j = 0, \dots, m-1$ capacity B and range $j \to m$. Then $E_H = m^2 B$, $E_C = Bm(m+1)/2$. At unit prices the threshold $L_0 = E_H - E_C = Bm(m-1)/2$ is attained only if all cold demands are met.

Suppose B attains this threshold. For each hot stream with positive heat, let r_i be the largest cold inlet index among its active partners; put $r_i = m$ if it is idle. Let $x_r = \sum_{i:r_i=r} a_i$. Boundary feasibility forces its hot outlet to be at least r_i, so total recovered heat is at most $\sum_{r<m}(m-r)x_r$. Consequently

$$L := \sum_{r=0}^{m} r x_r \le E_H - E_C = L_0. \tag{28}$$

For each $k = 1, \dots, m-1$, all cold streams with inlet at least k must draw from hot streams classified $r \ge k$. Their demand and an upper bound on their supply are

$$D_k = B \sum_{j=k}^{m-1} (m-j), \qquad S_k = \sum_{r=k}^{m-1} (m-r)x_r, \qquad S_k \ge D_k. \tag{29}$$

Use strictly positive multipliers

$$\lambda_k = \frac{k}{m-k} - \frac{k-1}{m-k+1} = \frac{m}{(m-k)(m-k+1)}\,. \tag{30}$$

Telescoping gives $\sum_k \lambda_k S_k = \sum_{r=1}^{m-1} r x_r$ and $\sum_k \lambda_k D_k = L_0$. Together with (28), these imply $x_m = 0$ and equality in every suffix constraint. Successive differences of (29) give $x_r = B$ for $r = 1, \dots, m-1$; total capacity gives $x_0 = B$. Thus the items are partitioned into m bins of capacity B and hence into triples. Conversely, assigning each partition group entirely to its cold stream attains the threshold and obeys the boundary approaches. The construction is strong and uses one stage.

These models really differ. For example hot $(F, H, \underline{H}) = (1, 10, 0)$ and cold streams $(G, C, \overline{C}) = (1, 8, 10), (1, 0, 8)$ have one-stage maximum recovery $42/5$ in C, but only 8 in B, at $\delta = 0$. For C, the LP inequalities $q_1/2 + q_2/10 \le 1$ and $q_2 \le 8$ give the sharp bound $q_1 + q_2 \le 42/5$, attained at $(2/5, 8)$. For B, positive heat to the high cold stream forces the mixed hot outlet to be at least 8, so total recovery is at most 2; omitting that match permits exactly 8. With two stages both attain 10, by traversing $10 \to 8 \to 0$.

6 Decision complexity and the encoding of the stage count

For fixed or unary ℓ, the projected B problem is a union of polynomial-size rational polyhedra: guess the set of permitted active matches, fix $q = 0$ off that set, and impose (9) on it. Zero heat on a guessed match is harmless. The remaining equations are (10), bounds and the cost threshold. A nonempty rational polyhedron has a rational feasible point with polynomial bit length, giving NP membership. The lifting lemma recovers the original branch variables without increasing that bound superpolynomially. The reductions therefore give strong NP-completeness.

For C, the active implications can instead be written as polynomial inequalities

$$q_{ijk}(h_{i,k-1} - b_{ijk} - \delta) \geq 0, \qquad q_{ijk}(a_{ijk} - c_{jk} - \delta) \geq 0. \tag{31}$$

Together with the bilinear equations, these give an existential formula over the reals of polynomial size. This proves membership in $\exists\mathbb{R}$ for fixed or unary stage counts. Bilinearity alone is not a proof of NP membership, and we do not claim that all feasible instances have small rational witnesses.

For completeness the following stage bound removes an exponential expansion when ℓ is encoded in binary. It uses the absence of forbidden matches and of extra design restrictions in our precise model.

Lemma 32 (Compression of the number of stages)**.** Any terminal mixed temperatures achievable by C or B with any number of stages are achievable by both models with at most $2(n + s) - 1$ stages.

Proof. Shift cold temperatures by δ to take $\delta = 0$; denote the attained hot ends by x_i and cold ends by y_j. Put $\phi_\theta(t) = \max(t - \theta, 0)$. Every feasible solution necessarily satisfies, for all real θ,

$$A(\theta) := \sum_i F_i[\phi_\theta(H_i) - \phi_\theta(x_i)] \;\geq\; D(\theta) := \sum_j G_j[\phi_\theta(y_j) - \phi_\theta(C_j)]. \tag{33}$$

To see this for C, an active exchanger has $a \geq c$ and $h \geq b$. The uniform points $(1 - u)a + uh$ dominate $(1 - u)c + ub$ for $0 \leq u \leq 1$. Since ϕ'_θ is nondecreasing, the average derivative on $[a, h]$ is at least that on $[c, b]$. Multiplying by $q = f(h-a) = g(b-c)$ gives $f[\phi_\theta(h) - \phi_\theta(a)] \geq g[\phi_\theta(b) - \phi_\theta(c)]$. The zero-heat case contributes zero. Convexity and the mixing equations add nonnegative Jensen gaps for both hot and cold mixing, exactly as in Lemma 17. Sum and telescope to obtain (33). For B use the isothermal lift first. Total recovered hot and cold heats are equal by energy conservation.

Sort the distinct numbers among H_i, x_i, y_j, C_j in decreasing order. There are at most $2(n + s)$ such numbers. Use one stage per consecutive pair $u > v$. Prescribe each cold stream's boundaries in that stage by clipping u, v to its range $[C_j, y_j]$. Thus any cold stream active in this interval has outlet u, inlet v, and demand $G_j(u - v)$.

Process the intervals downwards. Before the interval (v, u), a hot stream with $H_i \geq u$ has current temperature $p_i \geq \max(x_i, u)$, by induction; all other hot streams remain at their inlets. The available capacity to release heat in this interval is

$$F_i[p_i - \max(x_i, v)] \tag{34}$$

for the former streams, and zero for the latter. The total of these capacities is $A(v) - D(u)$: earlier intervals have consumed exactly $D(u)$, and streams not yet eligible have no supply above v because no inlet lies strictly between consecutive endpoints. By (33), it is at least $D(v) - D(u)$, the current cold demand. Choose any transportation of that demand within the available row capacities. Lower each hot temperature by its allocated heat divided by F_i. On each active match its previous temperature is at least u and its next temperature is at least v, so both boundary approaches hold. This also proves the induction invariant for the next interval.

At the bottom all cold streams have reached their required y_j. Equality of total hot and cold heat, and the lower bounds $p_i \geq x_i$, force every hot stream to reach x_i. Apply

the isothermal lift in each stage. The construction uses at most $2(n+s)-1$ stages and proves the claim. If no temperature interval exists, one idle stage suffices. □

Idle stages can always be inserted. Therefore, for the decision problem we may replace a binary-encoded ℓ by $\min\{\ell, 2(n+s)-1\}$. If ℓ is below this bound nothing changes; otherwise Lemma 32 preserves exactly the attainable terminal temperatures and costs. The NP and $\exists\mathbb{R}$ memberships above consequently hold for binary stage encoding too. This compression does not make two stages equivalent to an unrestricted number of stages: the explicit example above and the reductions retain their stage restrictions.

For these models minima are attained. Bound all boundary temperatures by their stream endpoints. All branch flows and heats are then bounded. In C, an active branch temperature lies between nonnegative stream inlets and a global maximum inlet/target temperature by the approaches; when $q = 0$, any positive flow has its inlet temperature, and a zero-flow temperature can be set to that inlet. Hence one may restrict all branch temperatures to a compact box. The polynomial implication formulation is closed. For B the bounded projected feasible set is a finite union of closed polyhedra and the lifting lemma applies. In particular, the zero-utility decision in the reduction is about an attained optimum, not an unattained infimum.

7 Exact checks, exploration and limitations

The proofs above stand independently of all computation. `scripts/verify_reduction.py` uses SymPy polynomial expansion and Python rational arithmetic to check the exchanger and mixing identities, the endpoint budgets, 36 full witnesses with stages 2–5 and approaches $0, 1, 3/2$, five explicit equality-case enumerations, and every sorted two-bin 3-Partition promise instance with $7 \le B \le 16$ (21 instances, 18 YES and 3 NO). All equations, nonnegativity and both active approach versions are checked on each saved witness. The negative finite enumerations use the equality lemma to reduce the feasible paths to a finite set; they are not independent nonlinear infeasibility certificates. The script also checks the one-stage boundary proof's multiplier identities for $2 \le m \le 12$. The JSON certificate contains exact fractions, not rounded solver output.

`scripts/verify_model_audit.py` retains the earlier exact audit, including the literal model construction, the arXiv sign defect, and the one-stage LP example. Its historical mode name `repaired` refers to the arXiv signs; those signs are already correct in the journal. `scripts/explore_physical.py` implements a floating-point SCIP MILP for B and a nonconvex quadratic model for C. The small stage-gap and initial one-stage experiments met their numerical expectations. The three new counter-current potential-gadget runs reached their 20-second time limits without settling the target values, including the YES instance; their logs explicitly say FAIL for an unresolved numerical expectation. They do not contradict or certify a theorem. Exact witnesses and the algebraic proof replace those solver inferences.

The result settles the polynomial-versus-NP-hard question for the two explicit physical completions, while the source's literal display is easy. It does not assert hardness for every possible engineering superstructure, for bounded numbers of streams, or for unrestricted stage counts chosen without a supplied stage budget. General NP membership of C remains unproved here; strong NP-hardness does not depend on that membership. The bounded novelty audit found no previous resolution, but its citation coverage was incomplete and several citing works were screened only by metadata

or abstracts. Detailed sources and access limitations are recorded in `references.md`. Independent verification of this preliminary argument is pending.

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URL: https://github.com/waikato-ahuora-smart-energy-systems/OpenHENS/blob/5081babe185e2b9c415970de7d269849ac02300b/openhens/classes/stage_wise_model.py (visited on 05/11/2026) 3

[5] Michael R. Garey and David S. Johnson. Computers and Intractability: A Guide to the Theory of NP-Completeness. Problem SP15, 3-Partition; strong NP-completeness. W. H. Freeman, **1979**. 5